



Artificial Intelligence

Project 4

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of an Artificial Intelligence Engineer at DecodeLabs. Project 4 is your Optional Mastery Phase: Image or Text Recognition (Basic). This track isn't about "complex training"—it's about Model Implementation. Since this is an optional milestone, you can choose to claim your certificate now or complete this task to prove you can integrate pre-trained AI libraries into a functional workflow. By finishing this, you demonstrate your ability to interpret and display model outputs through pure algorithmic logic.



Project 4: Image or Text Recognition (Basic)



Goal:

Implement a basic image or text recognition task using available libraries.

Key Requirements:

- Use a pre-trained model or simple library
- Perform recognition on sample input
- Display the output clearly

Key Skills:

Using AI libraries, understanding model outputs

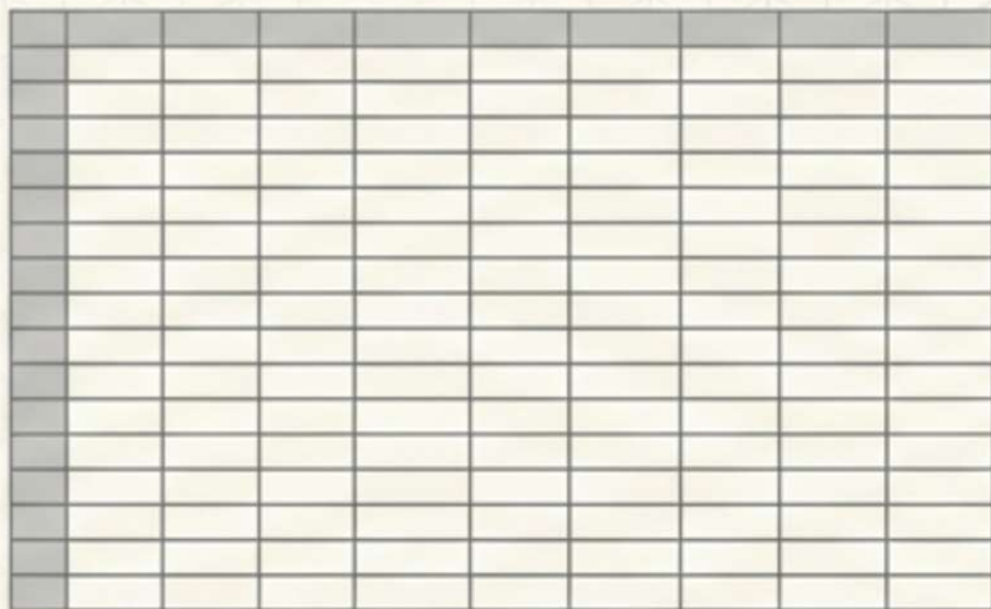
Project 4: Building the Machine's Optic Nerve

The DecodeLabs Architect's Playbook
for Image & Text Recognition

```
> EXECUTE_ONBOARDING_SEQUENCE // STATUS: READY
```


The Paradigm Shift: From Structured to Unstructured Data

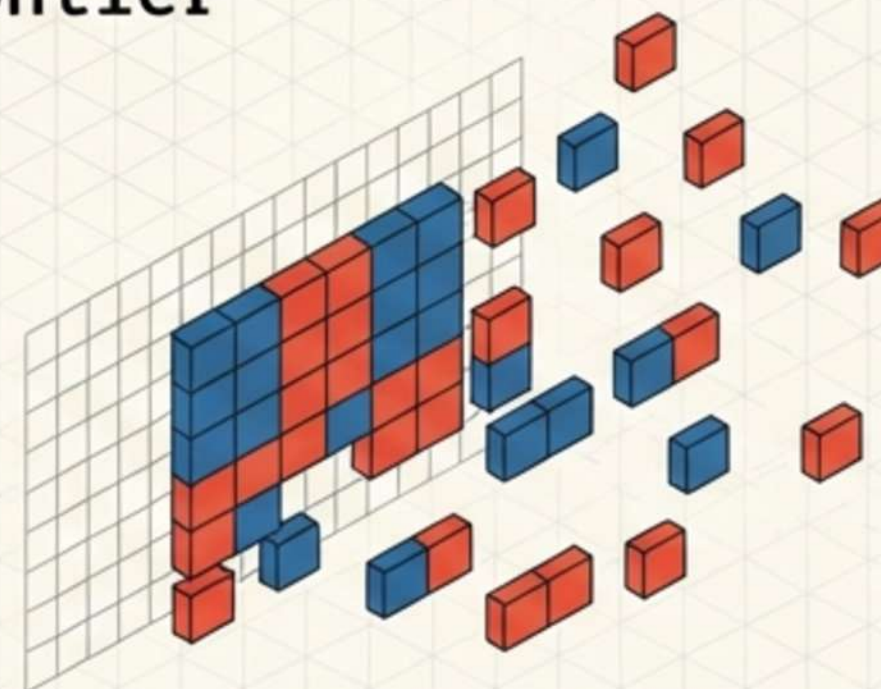
The Past



Structured Data: Spreadsheets, databases, clean CSVs.

The Limit: Accounts for < 20% of global enterprise data.

The Frontier



Unstructured Data: Scanned documents, video feeds, raw images.

The Opportunity: Over 80% of enterprise data lives here.

Mastering Project 4 means building the bridge between the physical world and computational logic. You are graduating to machine perception.

Project 4 Mission Parameters



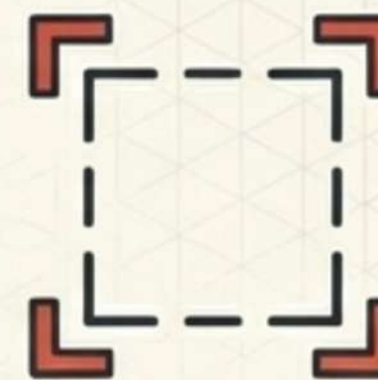
Objective

Engineer a Python script capable of ingesting raw visual data and extracting accurate, machine-readable intelligence.



The Toolkit

- pytesseract (Google's OCR Engine)
- OpenCV (Open Source Computer Vision)
- MobileNet-SSD (Deep Learning Architecture)

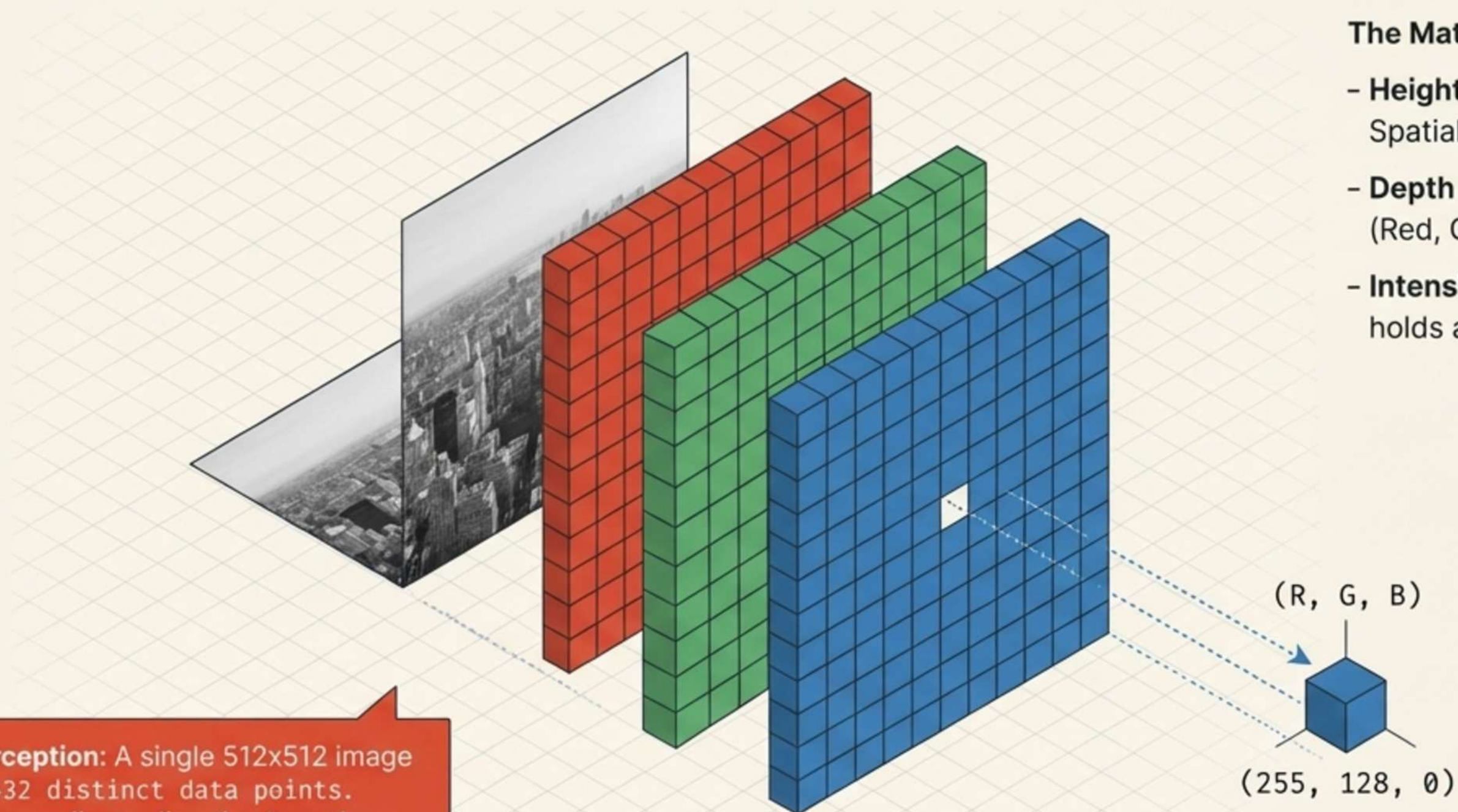


The Deliverable

A fully functioning recognition pipeline that proves the machine can see text or objects with validated confidence.

The IPO Model: Deconstructing the Visual Input

To a machine, an image is not a picture; it is a massive three-dimensional array.



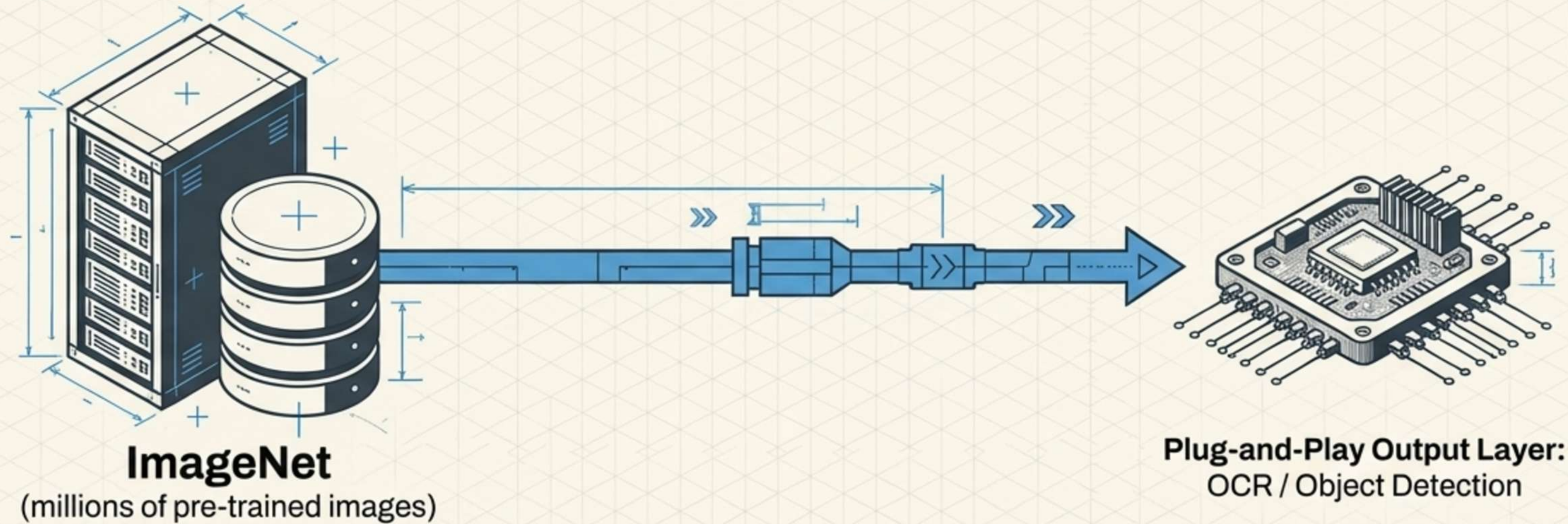
The Matrix Anatomy:

- **Height (H) & Width (W):** Spatial pixel resolution.
- **Depth (C):** 3 Color Channels (Red, Green, Blue).
- **Intensity:** Every pixel channel holds a value from 0 to 255.

The Scale of Perception: A single 512x512 image generates 786,432 distinct data points. Altering a single coordinate directly alters the machine's reality.

Transfer Learning: Inheriting the Machine's Knowledge

Why train an AI from scratch when you can download a degree?



How It Works

1. **The Base:** We leverage pre-trained models (like MobileNet) that have already analyzed millions of images to understand universal visual concepts (edges, shapes, gradients).
2. **The Transfer:** We detach the final output layer and plug in our specific task.

The Advantage

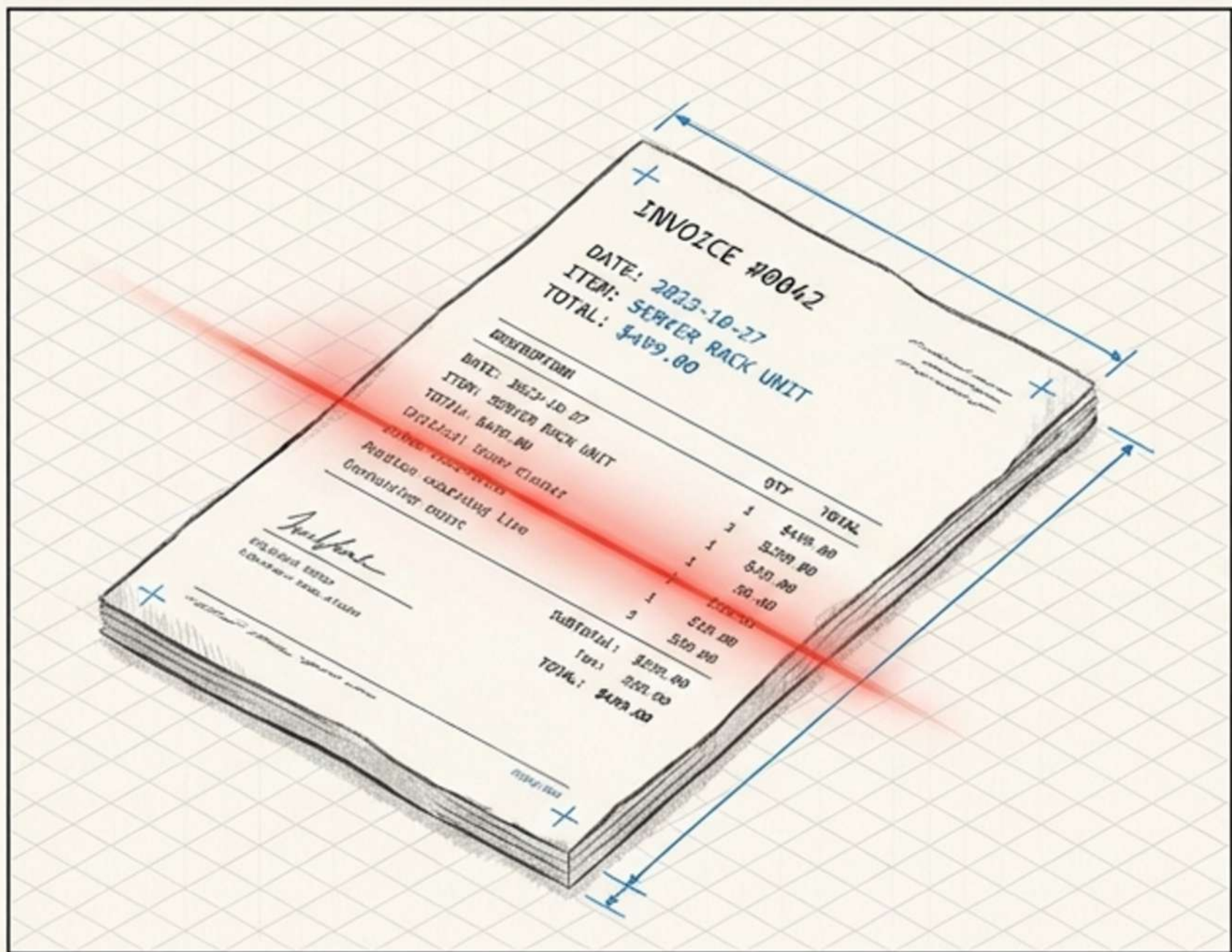
- Achieves high-accuracy perception using significantly less training data and local compute power.

The Perception Matrix: Choose Your Execution Path

	Path 1: OCR	Path 2: Object Detection
Core Objective	Extracting machine-readable strings	Identifying & locating physical entities
Primary Library	pytesseract	cv2.dnn & MobileNet-SSD
Pre-Processing	Grayscale, blur, adaptive thresholding	4D Blob construction (blobFromImage)
The Output	Formatted Text Strings	(X, Y, W, H) Bounding Box Coordinates

Path 1: Optical Character Recognition (OCR)

The Engine: pytesseract is our Python wrapper for Google's Tesseract engine, utilizing a convolutional + bi-directional LSTM pipeline to read sequences.



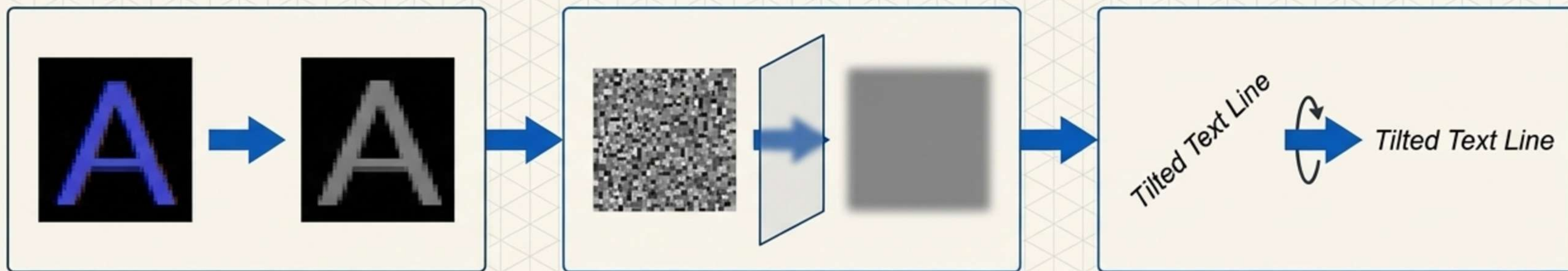
Tuning the PSM (Page Segmentation Mode)

- psm 3 : Fully automatic (Default for varied layouts).
- psm 6 : Single uniform block of text (Book pages).
- psm 7 : Single text line (Number plates/headers).
- psm 11: Sparse, scattered text (Invoices).

Layout configuration is critical for accuracy.

The Logic Skeleton: Systematic Image Pre-Processing

The Problem: Raw visual data is cluttered with shadows, chromatic noise, and uneven lighting.



Step 1: Grayscale Conversion

Collapses the 3D RGB matrix into a 1D intensity matrix. Removes distracting color data.

Step 2: Gaussian Blur

Smooths the image to eliminate micro-imperfections and artifact noise.

Step 3: Deskewing

Calculates **rotation** angles to snap tilted text back to a perfect horizontal baseline.

Adaptive Thresholding: Forcing the Binary Decision

The Mechanism: Thresholding forces every pixel to choose a side. It converts grayscale into pure black-and-white.



Input: Grayscale with shadows and noise.

Cutoff: 88



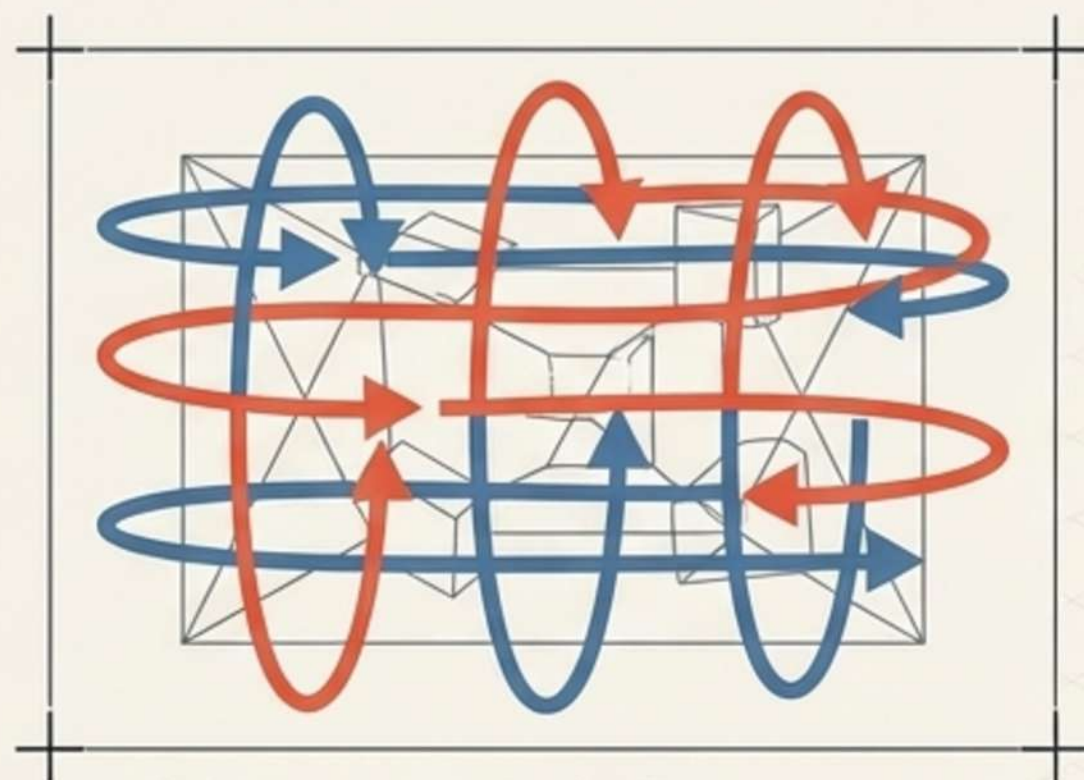
Output: Perfect contrast for character contours.

The Math (Otsu's Method):

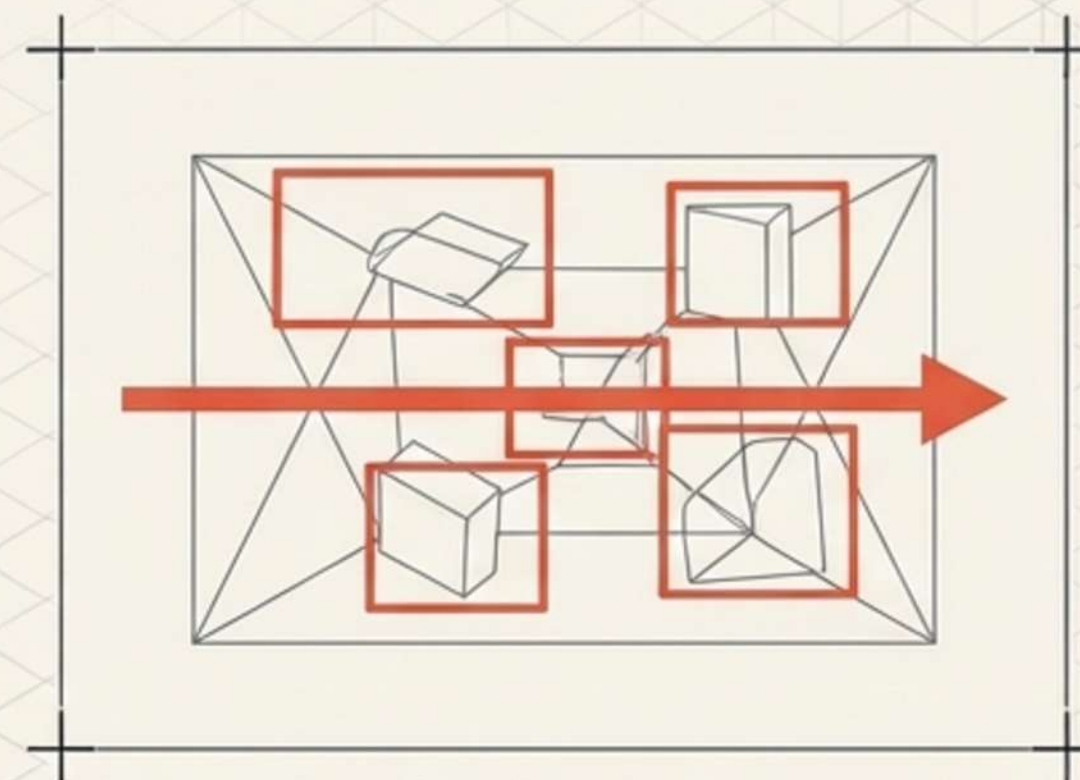
IF pixel_intensity $\geq$ 88 THEN pixel = 255 (White)

IF pixel_intensity $<$ 88 THEN pixel = 0 (Black)

Path 2: Object Detection with MobileNet-SSD



Old Way: Multiple Passes



SSD Way: Single Shot Detector

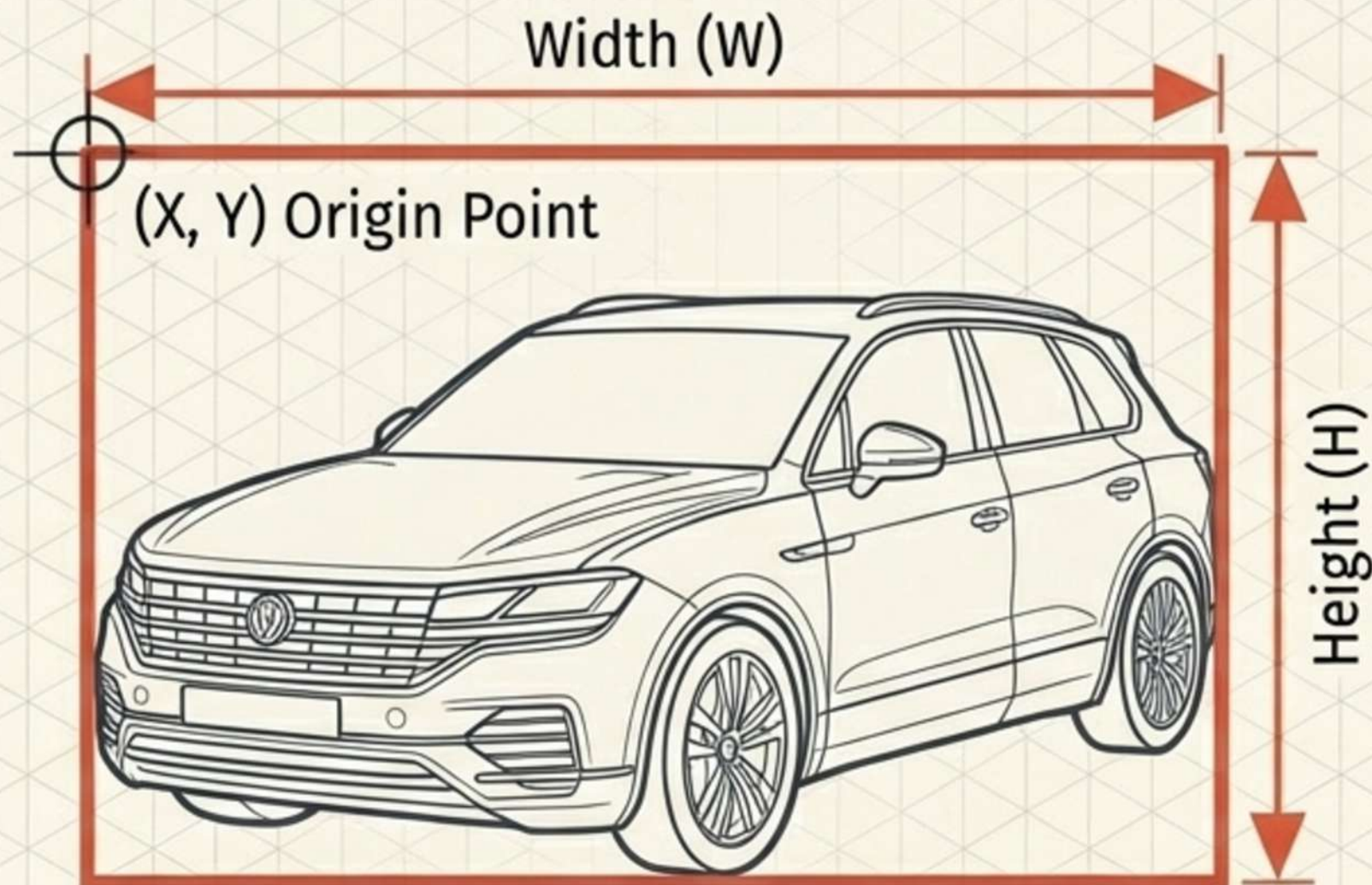
The Backbone: MobileNet v3

- Utilizes depthwise separable convolutions to filter input channels separately.
- Optimized for high-speed, real-time inference on edge devices with minimal compute requirements.

Step 1: Blob Construction

- We use `cv2.dnn.blobFromImage`.
- Performs mean subtraction.
- Scales the image to the required 300x300 network dimensions.

Decoding the Matrix: Anatomy of a Bounding Box



The Model Output:

The network doesn't output an image; it outputs normalized spatial coordinates.

Coordinate Scaling

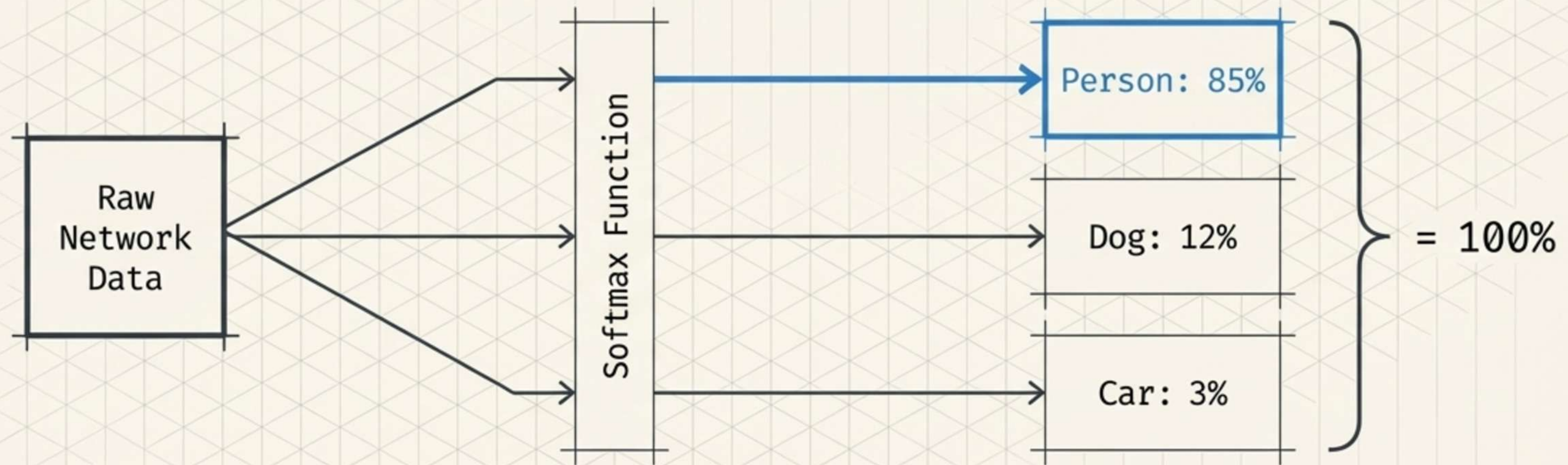
- Origin Point (X, Y): The top-left anchor of the detection.
- Dimensions (W, H): The calculated width and height of the entity.

The Translation:

We multiply these normalized coordinates by the actual pixel width and height of the original image to physically draw the bounding box overlay.

Decoding the Machine's Mind: Softmax & Confidence

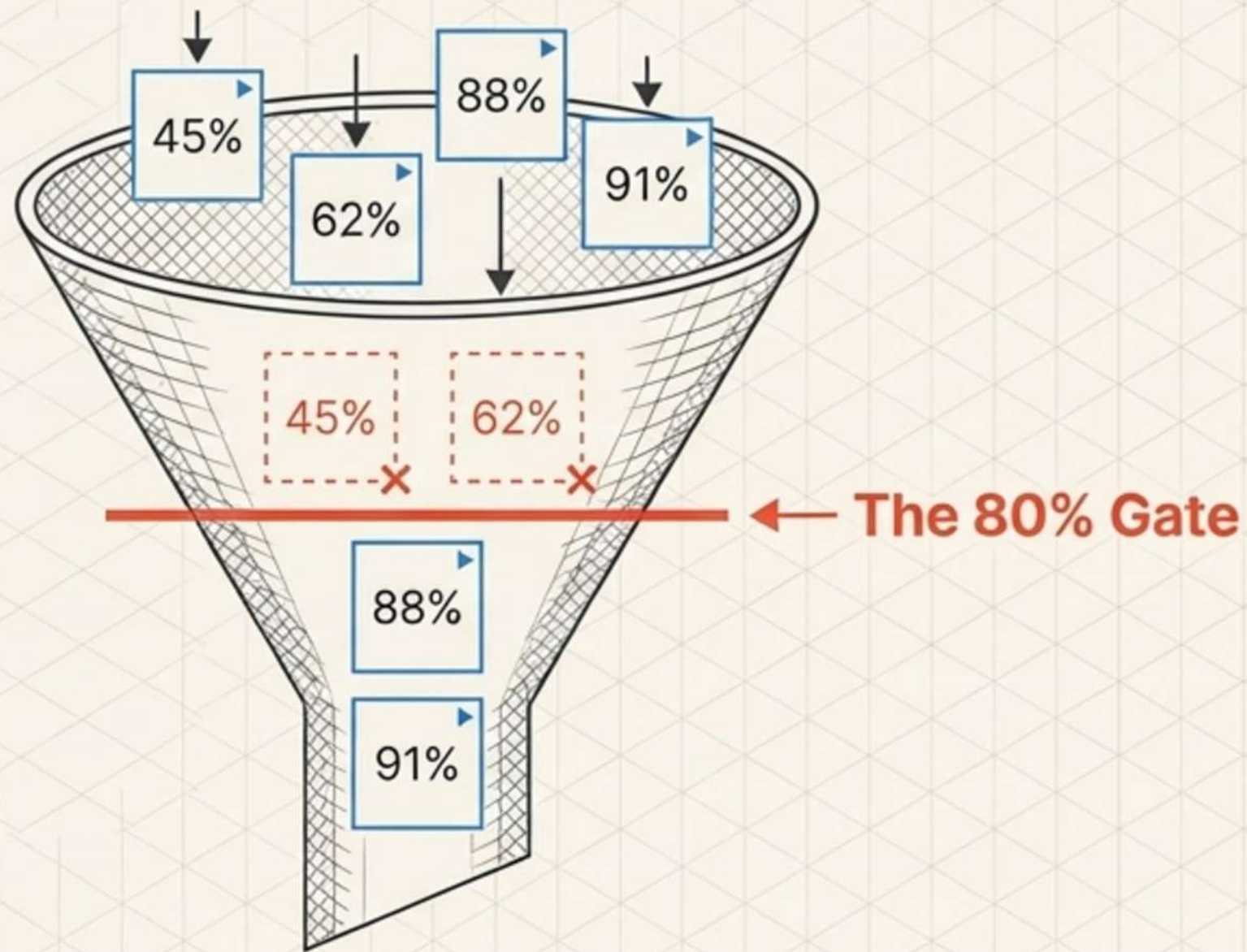
The Reality of AI: AI does not 'know' what an object is. It calculates the statistical probability of what an object might be.



Confidence Scores: Every bounding box or text string generated by the model comes with a confidence score attached. It is the machine's own assessment of its accuracy.

The 80% Threshold: The Confidence Filter

The Risk: Without a filter, an AI treats every guess with equal certainty, leading to confident hallucinations and false positives.



The IF Statement:

```
if confidence >= 0.80:  
    draw_box_and_label()  
else:  
    drop_detection()
```

The Balance: High thresholds minimize False Positives but increase the risk of False Negatives. In Project 4, 80% is the absolute minimum standard.

The Gatekeeper Rule: Milestone Validation

The Standard: To complete Project 4, your script must pass four uncompromising technical validations.

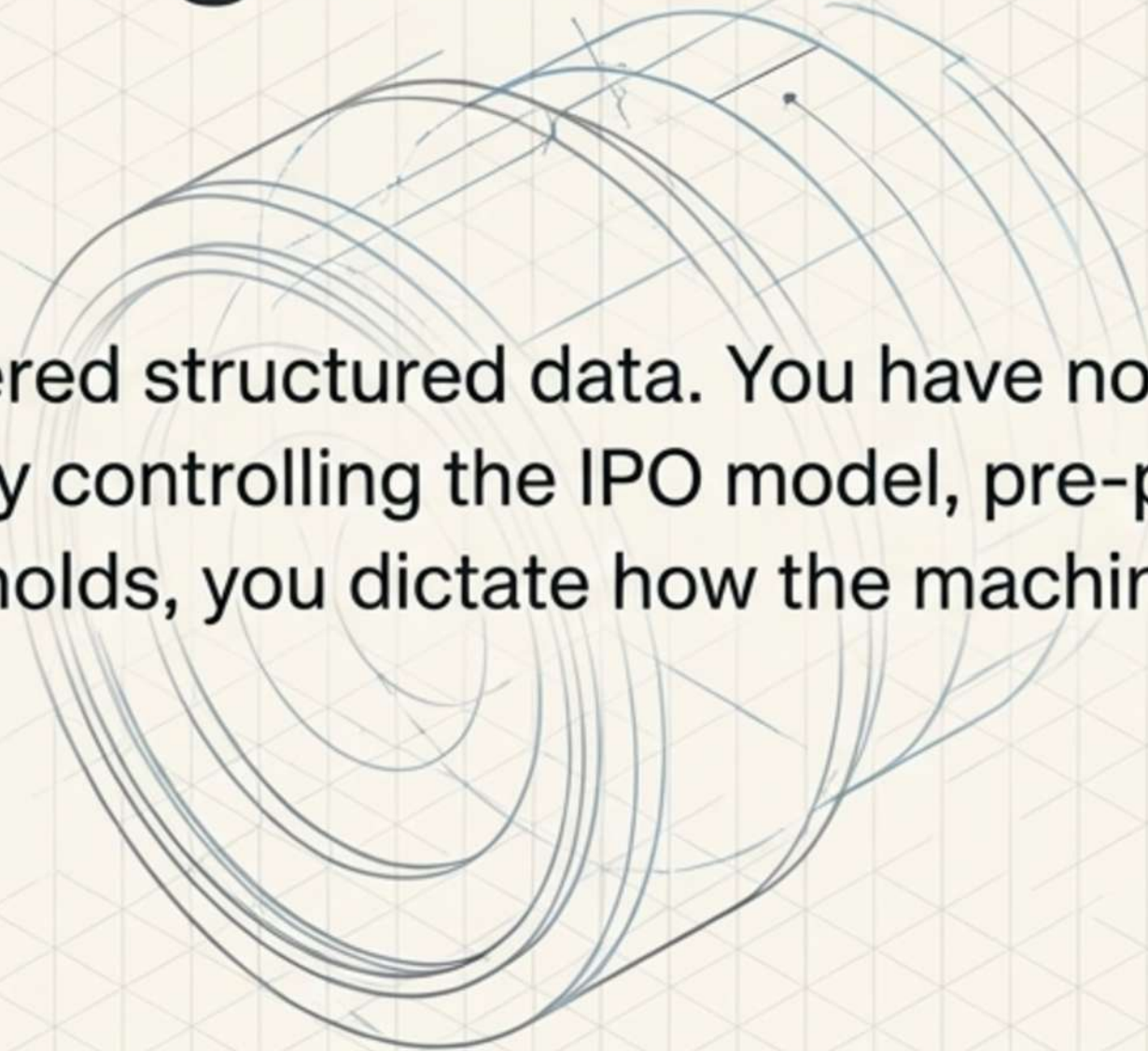
☐ **1. Library Integration**
Seamless, error-free implementation of pytesseract or cv2.dnn.

☐ **2. Pre-Processing Integrity**
Demonstrable execution of Grayscale conversion and Adaptive Thresholding to separate foreground from noise.

☐ **3. Accuracy Benchmarking**
A minimum validated confidence score of 80% on the final output.

☐ **4. Visual Confirmation**
Generation of a pristine visual output (legible OCR string or accurate bounding boxes with labels).

Architecting Machine Autonomy

An abstract sketch of a sphere or globe, composed of multiple overlapping, hand-drawn lines in shades of blue and grey. The lines create a sense of depth and movement, with some lines curving around the sphere and others intersecting it. The sketch is positioned in the center of the slide, behind the main text.

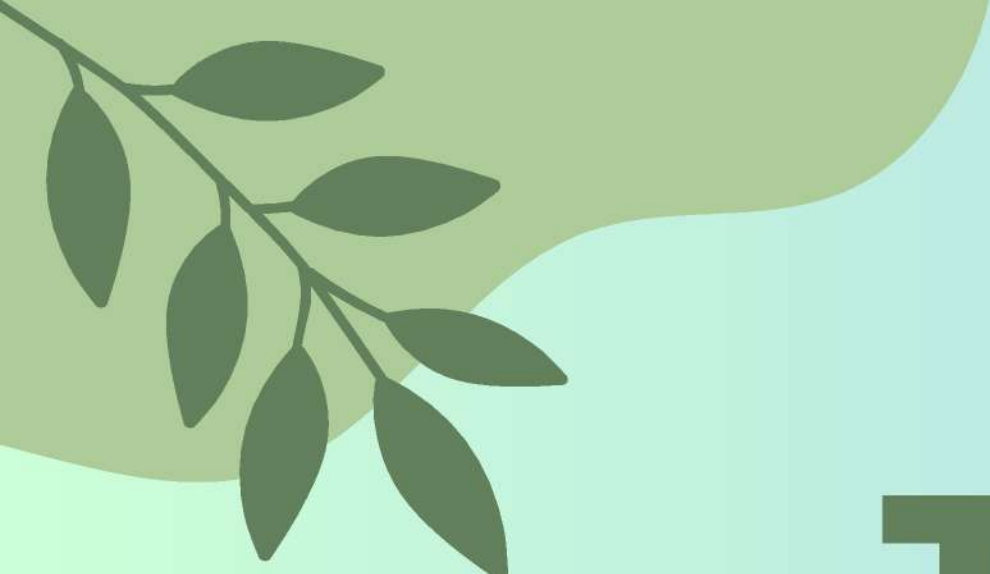
You have mastered structured data. You have now built the digital optic nerve. By controlling the IPO model, pre-processing, and confidence thresholds, you dictate how the machine perceives reality.

```
> INITIATE_PROJECT_4 // GOOD LUCK, ARCHITECTS.
```


CONCLUSION

The absolute best way to finish your time at DecodeLabs is by showcasing your ability to work with real-world AI applications. As this project is entirely optional, you have already met the core requirements for your certification! We are incredibly proud of the progress you have made throughout this journey.

Please submit all your completed tasks to the portal for final verification. We wish you the absolute best in your future career—keep innovating, keep learning, and keep building the future!



THANK YOU

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 GREATER LUCKNOW, INDIA

